

Gnasher Group:

An AI Dog Breed Group Finder

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ITAI 1378 Computer Vision and AI

Tier 1: Core Project

**Due September 26,
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THE PROBLEM

Many people can't identify their dog's breed group

Similar-looking breeds cause frequent confusion
like French Bulldogs vs. Pugs

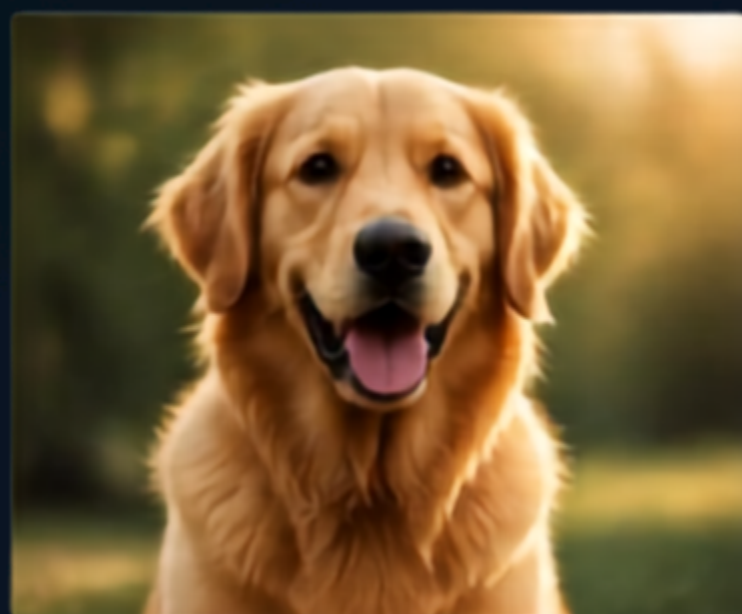
Veterinary experience shows the need
for a simple identification tool

Built for pet owners, adopters,
veterinary and shelter professionals



THE SOLUTION

Gnasher Group is an AI tool that identifies your dog's AKC breed group from a single photo.



DOG PHOTOGRAPH



IMAGE PREPARATION



EfficientNet-B0
MODEL

AI MODEL



HERDING
CONFIDENCE SCORE
93%

BREED GROUP + CONFIDENCE

AKC BREED GROUPS

SPORTING

HOUND

WORKING

TERRIER

TOY

NON-SPORTING

HERDING

TECHNICAL APPROACH

Image classification with
convolutional neural networks

EfficientNet-B0 using transfer
learning from ImageNet weights

Built with PyTorch and torchvision on
Google Colab

Efficient, lightweight, and ideal for
free Colab GPU use

Tier 1: single model for one
classification task



DATA PLAN

20,580

images from 120 breeds in Stanford Dogs

7

AKC breed groups for labeling

~7,000

balanced images, max 1,000 per group

70 / 15 / 15

training / validation / testing split



SUCCESS METRICS

Achieve **$\geq 85\%$** accuracy
on unseen test images

Average prediction time
 < 1 second per image on Colab GPU

Evaluate with a confusion matrix
and review **3–5** incorrect predictions



MILESTONE PLAN



RISKS AND RESOURCES

Similar breeds can confuse the model.

We show the top two predictions and alert when confidence is low.

Class imbalance affects accuracy.

We use dataset balancing, class weights, and augmentation to improve results.

Built on reliable, accessible tools.

Google Colab GPU, PyTorch, Stanford Dogs dataset, GitHub, and Google Drive.

Estimated cost: \$0

